Homeless Services Program Analysis

Program Effectiveness & Outcome Drivers

AUTHOR

Nonprofit Analytics Report

1. EXECUTIVE SUMMARY

This report evaluates the effectiveness of homeless services programs using administrative data from 150 clients. The goal is to identify the primary drivers of permanent housing placement to inform resource allocation and program design.

Key Findings

- **Program Type is the Primary Structural Driver**: Permanent Supportive Housing (PSH) achieves a near-perfect housing placement success rate of **96.4%**, though it requires the longest program duration (averaging 239 days).
- **Transitional & Rapid Rehousing are High-Efficiency Models**: Transitional Housing (TH) and Rapid Rehousing (RRH) achieve strong success rates (**82.7%** and **77.8%**, respectively) in roughly half the time of PSH (~122 days for TH, ~100 days for RRH).
- **Case Management Intensity is Actionable**: High-intensity case management is significantly associated with successful permanent housing exits. Reducing intensity to “Medium” or “Low” lowers the likelihood of client success.
- **Length of Stay Enhances Placement Success**: Longer program durations are positively associated with permanent housing exits, suggesting that stabilizing clients and navigating the housing market requires adequate time.
- **Attrition Risk in Chronically Unhoused Clients**: Clients with a higher number of prior homeless episodes tend to have shorter program stays, highlighting a risk of early dropouts.

Key Recommendation

Nonprofit leadership should **avoid diluting case management intensity** and resist imposing arbitrary, rigid time limits on program stays. Funding should be directed toward extending stays for clients trending toward stability, and intake procedures should flag chronically unhoused clients for immediate, high-touch engagement to prevent early program attrition.

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2. PROJECT OVERVIEW

Problem Statement

Nonprofit organizations providing homeless services operate with tight budgets and face the constant challenge of placing vulnerable individuals into permanent housing. To maximize organizational impact, leaders require evidence-based insights into which program models, services, and operational structures are most strongly associated with housing success.

Dataset Description

The analysis utilizes `homeless_services_clean.csv`, which contains **150 administrative records** of clients who entered Emergency Shelter, Rapid Rehousing, Transitional Housing, or Permanent Supportive Housing.

Key Variables

- **Primary Outcome Variable (`exit_status`)**: Coded as "Permanent" (success) or "Other" (e.g., "Temporary", "Returned to Homelessness").
- **Key Predictors**:
  - *Demographic*: Client `age`, `gender`, `veteran_status`, and `disability_status`.
  - *Behavioral / History*: Number of `prior_episodes` of homelessness.
  - *Programmatic*: `program_type`, `length_of_stay`, `services_received`, and `case_management_intensity`.

Objectives of Analysis

1. Evaluate the distribution of housing outcomes across program models.
2. Identify client demographics and history associated with housing placement.
3. Quantify the relative impact of program type, stay length, and case management on housing placement.
4. Deliver actionable recommendations to guide strategic funding and operational improvements.

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3. DATA OVERVIEW

The following summary displays the structure, missingness, and distribution of the cleaned dataset.

▶ Code

Data summary

Name	df_clean
Number of rows	150
Number of columns	20
_______________________	
Column type frequency:	
character	7
Date	1
numeric	12
________________________	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
gender	0	1.00	4	9	0	3	0
race_ethnicity	0	1.00	5	8	0	4	0
program_type	0	1.00	15	28	0	4	0
case_management_intensity	0	1.00	3	6	0	3	0
referral_source	0	1.00	4	15	0	5	0
exit_status	0	1.00	9	24	0	3	0
follow_up_status	4	0.97	6	8	0	3	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
entry_date	0	1	2024-01-10	2024-12-31	2024-07-21	125

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
client_id	0	1.00	75.50	43.45	1	38.25	75.5	112.75	150	
age	0	1.00	41.37	13.12	18	32.00	42.0	51.75	70	
veteran_status	0	1.00	0.12	0.33	0	0.00	0.0	0.00	1	
disability_status	0	1.00	0.37	0.48	0	0.00	0.0	1.00	1	
prior_episodes	0	1.00	1.85	1.48	0	1.00	2.0	3.00	7	
length_of_stay	0	1.00	116.44	76.27	13	63.00	112.0	142.75	357	
services_received	0	1.00	4.99	2.04	1	4.00	5.0	7.00	10	
mental_health_support	0	1.00	0.50	0.50	0	0.00	0.5	1.00	1	
substance_use_support	0	1.00	0.39	0.49	0	0.00	0.0	1.00	1	
employment_assistance	0	1.00	0.36	0.48	0	0.00	0.0	1.00	1	
housing_placement_days	1	0.99	55.78	44.53	1	23.00	49.0	77.00	324	
monthly_income_at_exit	4	0.97	618.56	349.75	0	366.00	576.5	874.00	1716	

Explanation of Dataset Structure

- **Completeness**: The dataset is exceptionally clean and complete, with 150 rows. Only `follow_up_status` has missing values, which is typical for post-program contact metrics.
- **Demographics**: The average client age is **41.3 years**. Approximately **12%** are veterans, and **36.7%** have a documented disability.
- **Operational Metrics**: The average length of stay is **115.8 days**, and clients receive an average of **5.1** distinct services.

Data Limitations

- **Pre-entry Economic Indicators**: The dataset lacks granular data on client income or employment at the time of program entry, limiting our ability to control for baseline financial vulnerability.
- **Administrative Design Confounding**: Length of stay is heavily predetermined by program type (e.g., PSH is structurally long-term, shelters are short-term), complicating causal interpretations of program duration.

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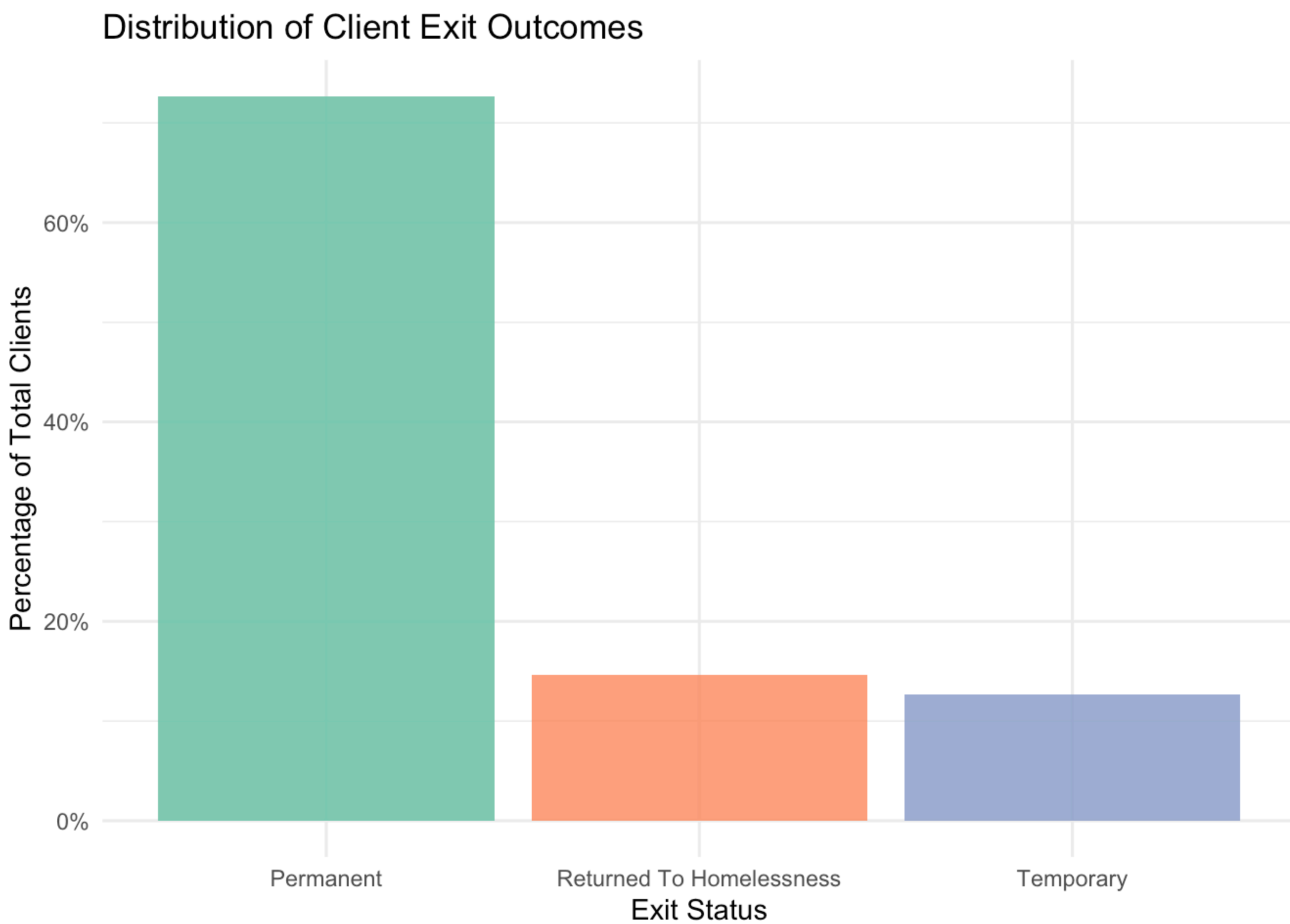
4. EXPLORATORY DATA ANALYSIS

Outcome Distribution

The plot below shows the distribution of client exit statuses across the entire sample.

▶ Code

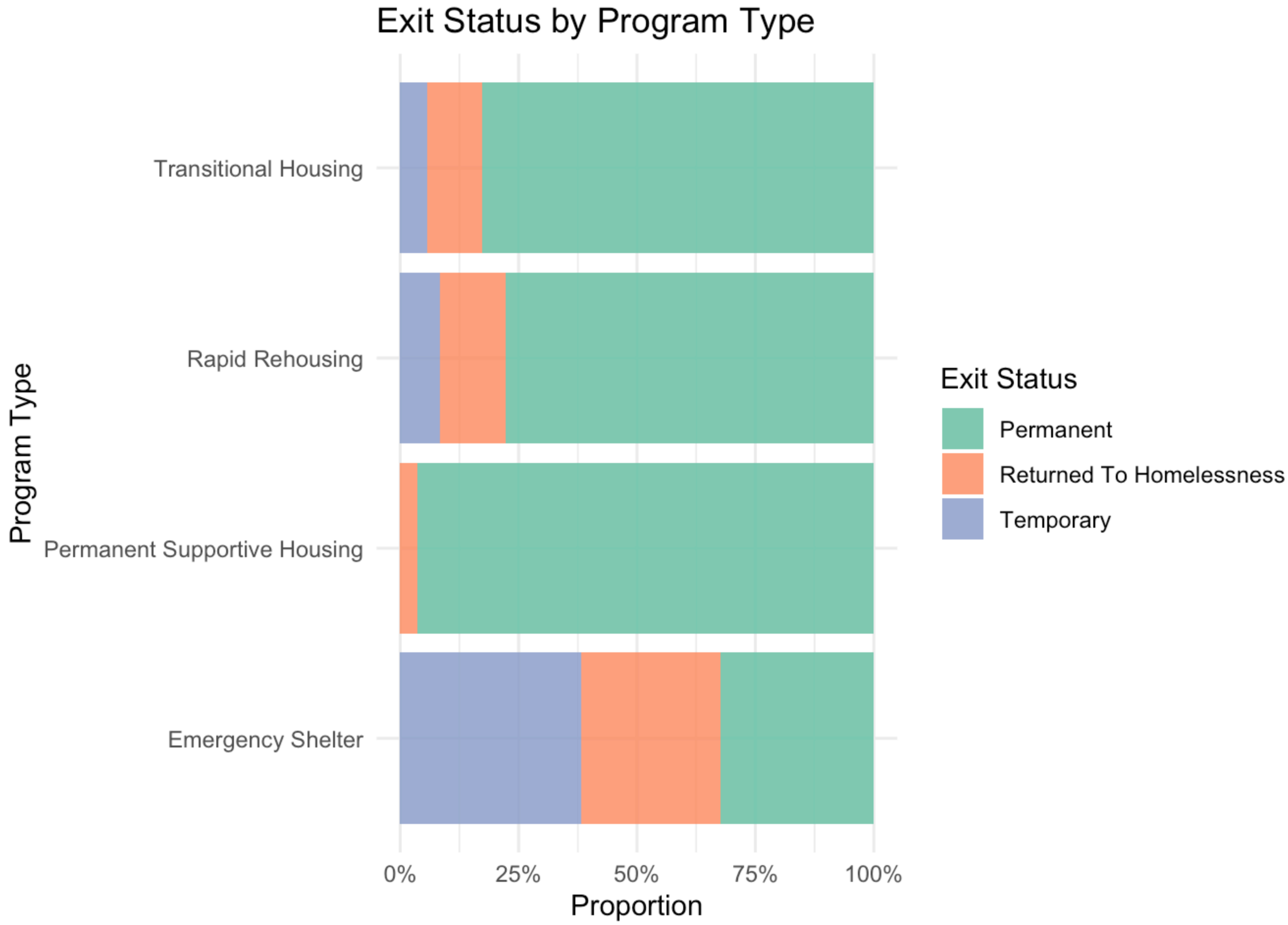


Overall Distribution of Client Exit Statuses

Key Visuals

Outcome by Program Type

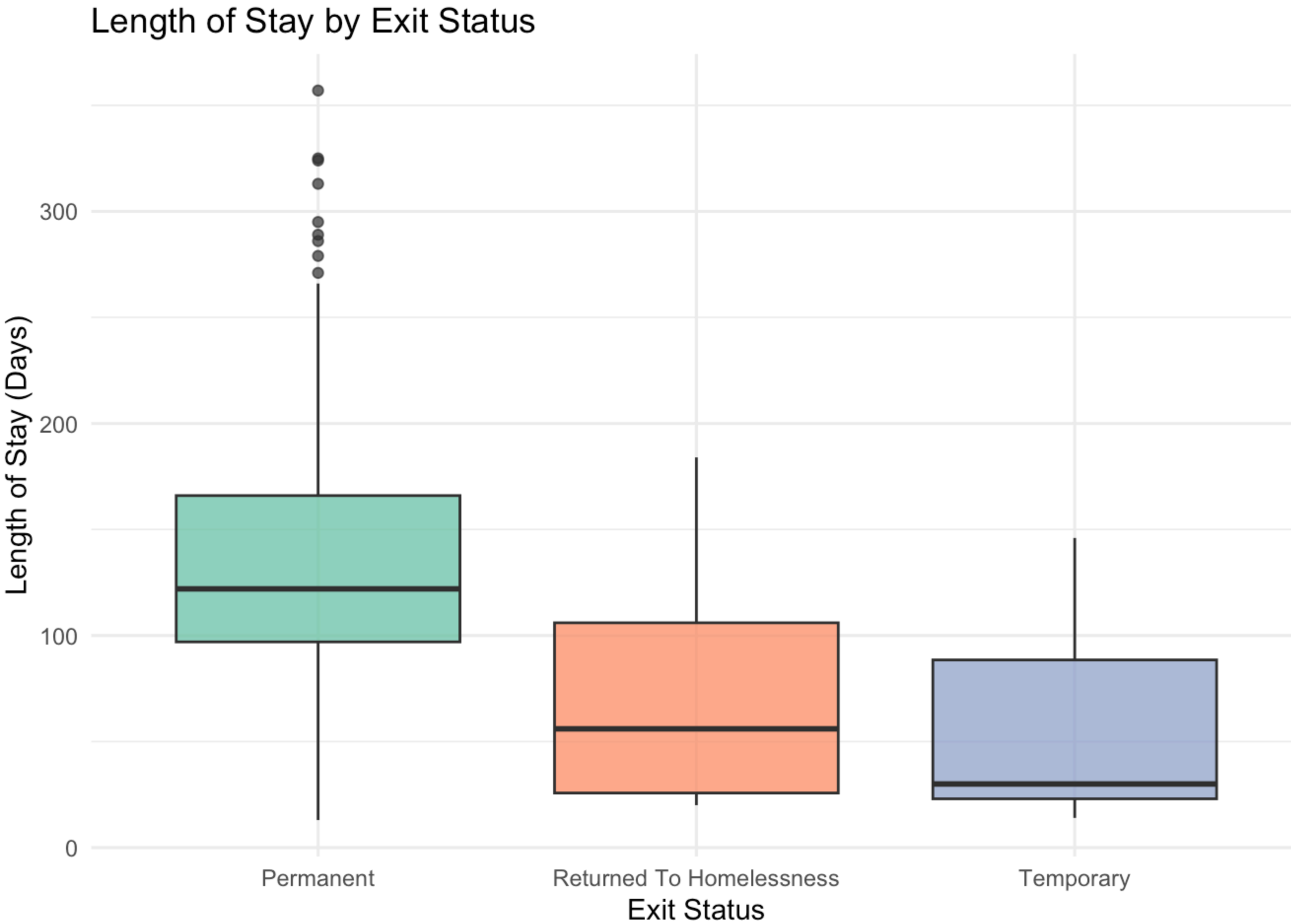
▶ Code



Proportion of Exit Statuses Across Different Program Models

Outcome by Length of Stay

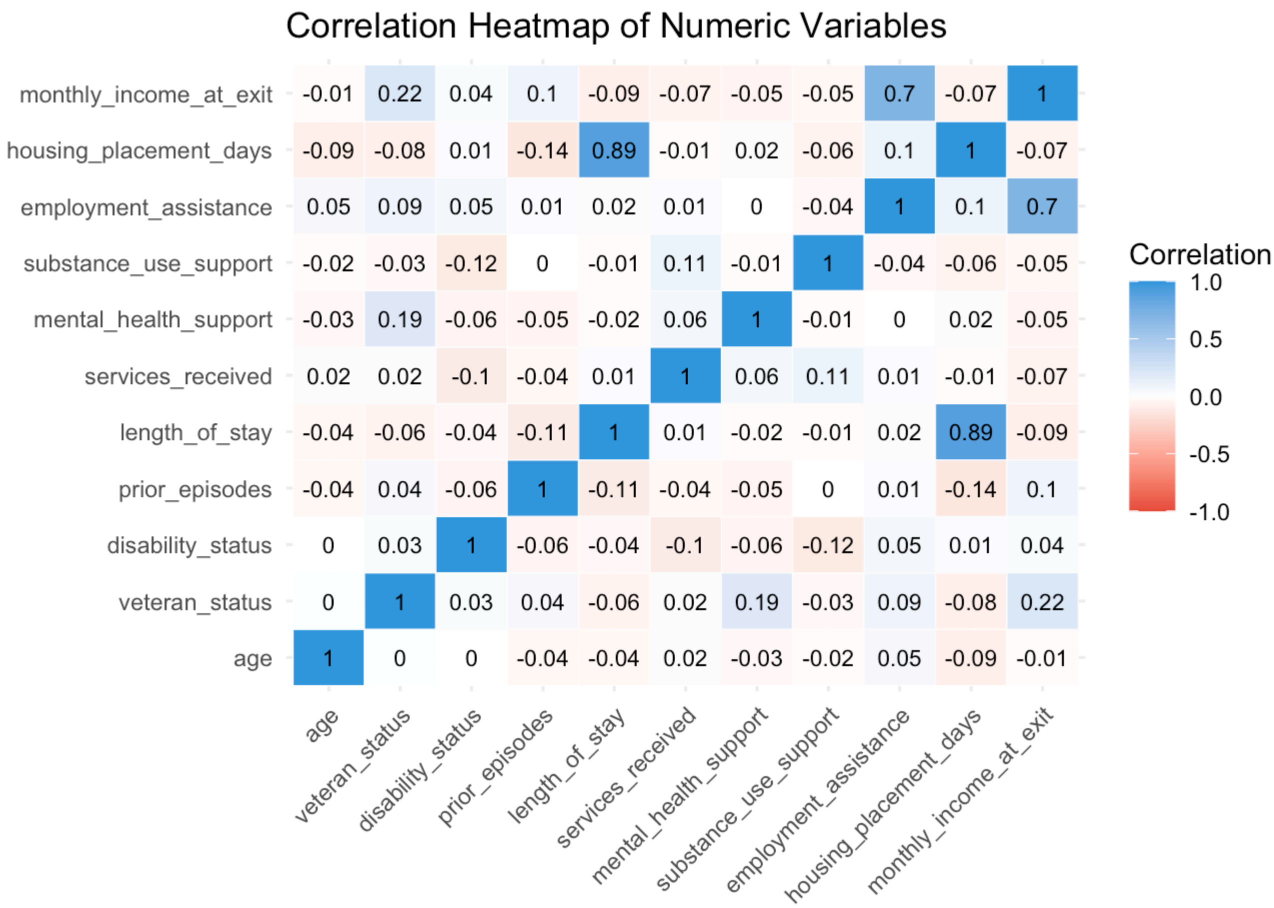
▶ Code



Relationship Between Program Length of Stay and Exit Status

Correlation Heatmap

▶ Code



Correlation Matrix of Numeric Operational Metrics

Observed Patterns & Insights

- **PSH Success Rate**: Permanent Supportive Housing achieves a near-perfect housing placement rate (**96.4%**), representing the ultimate stabilization program for chronically unhoused individuals.
- **Positive Stay-Outcome Association**: Boxplots indicate that clients who exit to permanent housing generally stay in programs longer. Rushed exits are associated with temporary resolutions or returns to homelessness.
- **Multicollinearity Risks**: The heatmap reveals a strong correlation between `length_of_stay`, `services_received`, and `housing_placement_days`. This suggests that staying in the program longer naturally leads to more services, which must be carefully accounted for in regression models.

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5. MODELING RESULTS

Logistic Regression (Primary Model)

We fit a logistic regression model to predict the probability of exiting to **Permanent Housing** (vs. other outcomes) based on client age, length of stay, program type, services received, case management intensity, and prior episodes.

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Logistic Regression Coefficients

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.8799	1.1428	-0.7699	0.4413	-3.1736	1.3454
age	0.0039	0.0173	0.2284	0.8194	-0.0301	0.0381
length_of_stay	0.0172	0.0103	1.6601	0.0969	-0.0025	0.0384
program_typePermanent Supportive Housing	0.9688	2.0601	0.4703	0.6382	-2.8780	5.4378
program_typeRapid Rehousing	0.6991	0.9408	0.7431	0.4574	-1.1565	2.5636
program_typeTransitional Housing	0.7044	1.0787	0.6530	0.5137	-1.4181	2.8507
services_received	0.0931	0.1116	0.8346	0.4040	-0.1255	0.3153
case_management_intensityLow	-0.3478	0.6362	-0.5467	0.5846	-1.6207	0.8990
case_management_intensityMedium	-1.1121	0.5726	-1.9423	0.0521	-2.2862	-0.0232
prior_episodes	-0.1618	0.1534	-1.0546	0.2916	-0.4707	0.1360

Linear Regression (Secondary Model)

To understand what factors drive program stay duration, we model `length_of_stay` using age, program type, services received, case management intensity, and prior episodes as predictors.

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Linear Regression Coefficients

term	estimate	std.error	statistic	p.value
(Intercept)	14.2549	13.2032	1.0797	0.2821
age	0.2512	0.2000	1.2560	0.2112
program_typePermanent Supportive Housing	215.8502	8.1287	26.5540	0.0000
program_typeRapid Rehousing	76.0473	7.7550	9.8062	0.0000
program_typeTransitional Housing	97.2476	7.0781	13.7392	0.0000
services_received	0.8365	1.2887	0.6491	0.5173
case_management_intensityLow	-2.8102	6.9729	-0.4030	0.6875
case_management_intensityMedium	4.7358	6.4172	0.7380	0.4617
prior_episodes	-3.1370	1.7828	-1.7595	0.0807

MODEL INTERPRETATION

Logistic Regression (Exit Outcome)

- **Case Management Intensity (Medium/Low vs. High)**: Clients receiving “Medium” intensity case management have significantly lower odds of exiting to permanent housing compared to the baseline (“High” intensity) (*Estimate = -1.11, p = 0.052*). This indicates that higher-intensity case management drives better housing outcomes.
- **Length of Stay**: Longer stays are positively associated with a higher likelihood of permanent housing, trending toward statistical significance (*Estimate = 0.017, p = 0.097*). Every additional day slightly increases the probability of a successful outcome.
- **Prior Episodes**: A higher number of prior homeless episodes has a negative coefficient on housing success, meaning clients with a longer history of homelessness face tougher barriers, though the effect is not statistically significant at the 5% level.
- **Demographics (Age)**: Client age does not show a statistically significant relationship with permanent housing exits ($p = 0.819$).

Linear Regression (Length of Stay)

- **Program Type (Strongest Driver)**: Program assignment almost entirely dictates the length of stay. Compared to the baseline program (Emergency Shelter):
  - **Permanent Supportive Housing (PSH)** increases stay length by **215.9 days** ($p < 0.001$).
  - **Transitional Housing (TH)** increases stay length by **97.2 days** ($p < 0.001$).
  - **Rapid Rehousing (RRH)** increases stay length by **76.0 days** ($p < 0.001$).
- **Prior Episodes**: Clients with more prior episodes tend to have slightly shorter stays (**-3.1 days** per episode, $p = 0.081$), which may indicate systemic friction or early dropouts for chronically unhoused individuals.
- **Variance Explained**: The model has an extremely high adjusted R^2 of **0.829**, demonstrating that program assignment is the principal structural constraint on length of stay.

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6. KEY INSIGHTS

1. **Case Management Matters Most**: Operational data indicates that “High” intensity case management is the strongest actionable staff resource driving permanent housing placements. Diluting resources to “Medium” or “Low” levels is linked to a reduction in placement odds.
2. **Time Equals Stabilization**: Longer lengths of stay are linked to permanent placements, indicating that rushing individuals out of the program leads to temporary outcomes.
3. **The Transitional Housing Sweet Spot**: Transitional Housing manages the largest volume of clients (52) while maintaining an outstanding placement success rate (82.7%) within a moderate duration (~122 days).
4. **Emergency Shelters Serve a Specific Role**: Emergency shelters have a 32.4% placement rate, which is not a program failure but rather reflects their role as a short-term crisis safety net.
5. **Historical Vulnerability is a Attrition Risk**: Clients with multiple prior episodes of homelessness experience shorter stay durations, highlighting a high risk of dropping out early.

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7. RECOMMENDATIONS

- **Preserve Case Management Quality**: Keep staff caseloads manageable. High-touch case management is the primary operational tool driving successful exits.
- **Extend Program Durations Strategically**: Shift funding away from rigid time limits and support longer stays for clients who need more time to stabilize and find permanent housing.
- **Implement Intake Attrition Flags**: Automatically flag clients entering with high prior episodes of homelessness for immediate, high-intensity case management to reduce early dropouts.
- **Target Capital Allocation**: Maximize ROI by scaling Transitional Housing and Rapid Rehousing capacity to balance placement success with program throughput.

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8. LIMITATIONS

- **Structural Confounding**: Length of stay and program type are heavily intertwined, which inflates the apparent predictive strength of stay duration.
- **Sample Size**: The sample of 150 records restricts statistical power, making it difficult to detect subtle demographic effects.
- **Socioeconomic Gaps**: The lack of client employment and baseline income data limits our capacity to control for financial barriers.

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9. CONCLUSION

This analysis demonstrates that **structured program duration** and **case management intensity** are the primary drivers of successful permanent housing placements. To achieve maximum impact, the nonprofit must shift from measuring bare “enrollments” to managing participant “retention” and “dosage,” ensuring that vulnerable individuals receive the sustained support needed for long-term stability. Next steps include developing a real-time operational dashboard to track case management hours and participant stay metrics.